

3 The treatment of many diseases has been improved by the use of genetic technology in medicine.

(a) Explain what is meant by genetic engineering.

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(b) (i) Leber Congenital Amaurosis (LCA) is an inherited eye disease. In LCA, the photoreceptor cells in the retina die at an early age. This causes impaired vision (reduced eyesight) in children, which can progress to blindness.

Mutations in different genes cause different forms of LCA. One form of this disease, LCA2, is caused by a mutation in the *RPE65* gene. Gene therapy has been used to treat LCA2.

Outline how an inherited eye disease, such as LCA2, is treated with gene therapy.

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(ii) Suggest why the eye is a suitable organ for gene therapy.

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- (c) LCA10 is a different form of LCA caused by a recessive mutation in the *CEP290* gene. This gene codes for the protein CEP290, which is involved in the correct functioning of photoreceptor cells in the retina.

The mutation in *CEP290* causes an error to be made when the primary transcript is spliced to form messenger RNA (mRNA). The abnormal mRNA that is formed has an extra sequence of RNA nucleotides, known as exon X, between exon 26 and exon 27. Exon X contains a STOP codon.

Fig. 3.1 compares the effect of the mutation in *CEP290* with the normal gene expression.

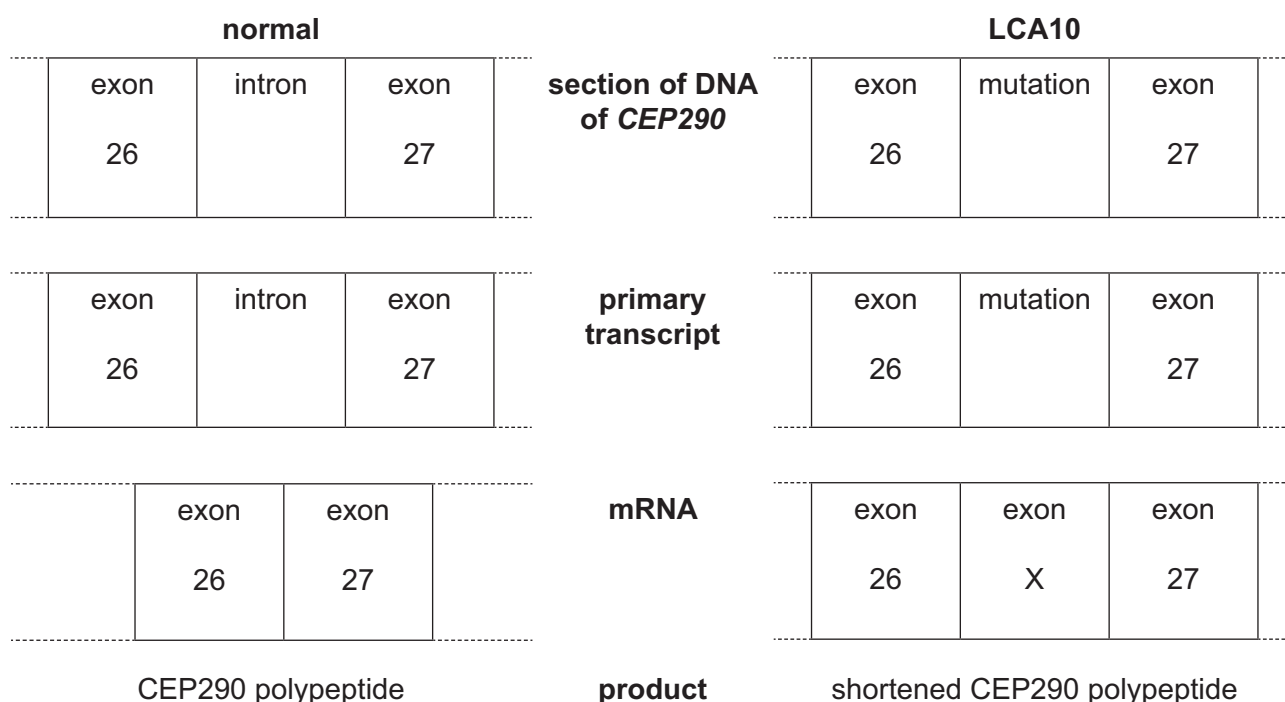


Fig. 3.1

In 2022, research was carried out into possible treatment of LCA10 using genetic technology.

- (i) A human clinical trial investigated a treatment of LCA10 using a short RNA nucleotide sequence known as Sepofarsen.

Sepofarsen binds to the section of the primary transcript containing the *CEP290* mutation so that normal splicing occurs and functional CEP290 protein is synthesised.

- People in the clinical trial received regular treatment with Sepofarsen to the eye with the greatest loss of vision (treated eye) for a period of 12 months.
- Changes in the light perception (visual acuity) of both eyes were measured using a vision chart.
- A negative change in the visual acuity score shows an improvement in visual acuity.

Fig. 3.2 shows the results of the clinical trial over 12 months.

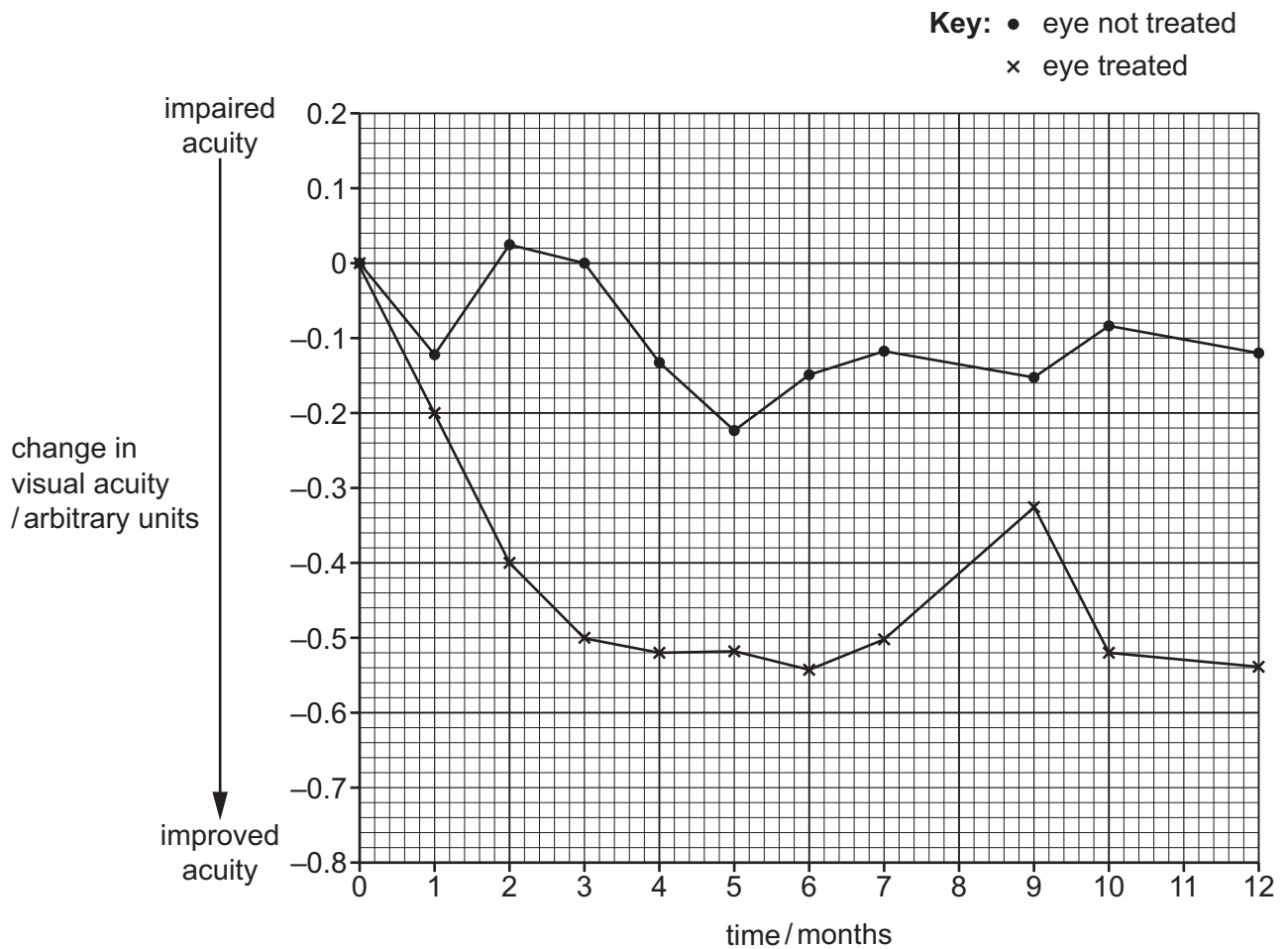


Fig. 3.2

Describe the results of the clinical trial data shown in Fig. 3.2.

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The cell repair mechanisms repair the cut in DNA after the modification has taken place.

- Explain how this method used to treat LCA10 is an example of gene editing.

[Total: 13]



Question 4 starts on page 10